AI PIPELINE

The 6-Agent System

A sequential triage chain followed by two feedback intelligence agents. Each triage agent builds on the last; the feedback agents learn across runs.

01 Enrichment | 02 Risk Assessment | 03 Categorisation | 04 Summary | 00 Auto-Label

Pure Logic (No LLM) Claude Haiku Claude Sonnet Claude Sonnet (batch)

AGENT 01 No LLM

Data Enrichment

The foundation of the pipeline. Before any AI model is invoked, Agent 01 performs pure deterministic data enrichment using in-memory index lookups. It assembles the complete context object that all downstream agents depend on. When a pre-attached session trail is available (set at alert generation time), it is included directly.

Transaction Record | Customer Profile | Account Details | 90-Day Session Trail

- 30-day transaction velocity — count and total volume
- Currency diversity score — number of distinct currencies
- Cash intensity ratio — cash transactions as % of total
- Days since last account activity
- Customer risk rating and PEP / deceased status
- Cross-account session trail (all accounts, 90-day window)

INPUT

Alert ID + Transaction ID + optional pre-attached session trail

OUTPUT

Enriched context object with computed rolling statistics and session context

AGENT 02 Claude Haiku

Risk Assessment

The first LLM agent. Claude Haiku receives the enriched context and evaluates the alert across five independent risk dimensions. It reasons about each dimension separately before identifying which Key Risk Indicators are most significant.

Transaction Risk Amount, type, frequency, structuring patterns

Customer Risk Profile, PEP status, occupation, risk rating

Account Risk Age, status, dormancy, currency mix

Behavioural Risk Velocity changes, unusual counterparties

Geographic Risk Jurisdiction flags, cross-border patterns

INPUT	OUTPUT
Enriched context object from Agent 01	5 risk dimension scores + ranked list of Key Risk Indicators (KRIs)

AGENT 03 Claude Haiku

AML Typology Categorisation

The classification agent. Claude Haiku takes the risk scores and KRIs from Agent 02 and maps the alert to the most appropriate AML typology from a defined taxonomy. It also assigns the final risk tier.

🔄 Structuring | 🐓 Layering | 💣 Terrorist Financing | 🥷 Fraud | 📊 Tax Evasion

💼 Bribery / Corruption | 🚫 Sanctions Evasion | ✅ Low Risk

🔴 CRITICAL | 🟠 HIGH | 🟡 MEDIUM | 🟢 LOW

INPUT	OUTPUT
Risk dimension scores + KRI list from Agent 02	AML typology classification + final risk tier

AGENT 04 Claude Sonnet



Investigation Summary

The most capable model in the triage chain. Claude Sonnet receives all outputs from the three preceding agents and synthesises them into a natural-language investigation brief. This is what the analyst reads. It is written to be immediately actionable.

- ✓ Plain-English narrative of why this alert was flagged
- ✓ The most significant KRIs and what they indicate
- ✓ AML typology explanation in plain language
- ✓ Risk tier justification
- ✓ Suggested next steps for the analyst
- ✓ Relevant contextual factors that may explain or escalate risk

AGENT 00 No LLM

Deterministic Auto-Labeler

Runs after triage completes — no LLM call required. Scores each completed alert's FP probability from its own triage signals using a fixed weighted formula. Provides an instant pre-classification to help analysts prioritise the queue.

- **LOW_RISK typology** +0.50 FP probability signal
- **CLOSE recommended action** +0.30 FP probability signal
- **Low classification confidence (<0.4)** +0.20 FP probability signal
- **Summary confidence LOW** +0.10 FP probability signal

`AUTO_FP ≥ 0.65` `AUTO_UNCERTAIN 0.26–0.64` `AUTO_TP ≤ 0.25`

INPUT

Completed alert with triage_result (categorisation + investigation_summary)

OUTPUT

predicted_verdict · predicted_fp_probability · predicted_reason (persisted to DB)

AGENT 05 Claude Sonnet

Evolution Intelligence

Operates on the full corpus of analyst-labeled alerts — not just a single alert. Claude Sonnet analyses patterns in false positives across all rules and typologies, then recommends targeted rule recalibration. Every run is persisted: patterns accumulate a confidence score based on how often they recur and how many FPs they explain.

- ✓ Detects which rules produce the most false positives
- ✓ Identifies recurring FP patterns (e.g. RULE_HV + LOW_RISK typology)
- ✓ Assigns P0 / P1 / P2 priority codes to each recommendation
- ✓ Pattern confidence score increases with each recurrence
- ✓ Tracks FP rate trend across all evolution runs
- ✓ What-if simulator: model impact of suppressing a rule before applying
- ✓ Auto-applies high-confidence, low-risk recommendations (configurable)

INPUT

All alerts with analyst verdicts (FALSE_POSITIVE / TRUE_POSITIVE / INCONCLUSIVE)

OUTPUT

FP patterns + P-code recommendations + trend data (all persisted to evolution_runs, evolution_patterns, evolution_recommendations)

DESIGN RATIONALE

Why This Architecture?

Separation of Concerns

Each agent is optimised for one task. Haiku handles fast scoring cheaply; Sonnet handles nuanced prose. Agent 0 never calls an LLM — it's always instant. Agent 5 operates on the full labeled corpus, not per-alert.

Cost Efficiency

The two Haiku agents handle heavy per-alert lifting cheaply. Sonnet is invoked once per alert (summary) and once per evolution run (batch analysis). Auto-labeling and session trail attachment require zero LLM calls.

Learning Loop

Agent 5 closes the feedback loop. The system improves with every analyst verdict — patterns build confidence over multiple runs, and recommendations get priority codes based on how much FP reduction they would deliver.

Auditability

Every step produces a discrete, inspectable output persisted to SQLite. Analysts see exactly what each agent concluded. Evolution runs have full provenance — which alerts, which patterns, which recommendations, when applied.